

AOS-CX 10.06 Link Aggregation Guide

6100, 6200, 6300, 6400, 8320, 8325, 8360, 8400 Switch Series



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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)
- Aruba 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A)
- Aruba 6300 Switch Series (JL658A, JL659A, JL660A, JL661A, JL662A, JL663A, JL664A, JL665A, JL666A, JL667A, JL668A, JL762A)
- Aruba 6400 Switch Series (JL741A, R0X26A, R0X27A, R0X29A, R0X30A)
- Aruba 8320 Switch Series (JL479A, JL579A, JL581A)
- Aruba 8325 Switch Series (JL624A, JL625A, JL626A, JL627A)
- Aruba 8360 Switch Series (JL700A, JL701A, JL702A, JL703A, JL706A, JL707A, JL708A, JL709A, JL710A, JL711A)
- Aruba 8400 Switch Series (JL375A, JL376A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and other resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
<code>example-text</code>	In code and screen examples, indicates text entered by a user.

Table Continued

Convention	Usage
Any of the following: <code><example-text></code> <code><example-text></code> <i>example-text</i> <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: - For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. - For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or	Ellipsis:
...	- In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. - In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment. The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

member/slot/port

On the 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 on the switch.

On the 6200 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 in slot 1 on member 1.

On the 6300 Switch Series

- *member*: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 10. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 on member 1.

On the 6400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.

- Management modules are on the front of the switch in slots 1/1 and 1/2.
- Line modules are on the front of the switch starting in slot 1/3.
- *port*: Physical number of a port on a line module.

For example, the logical interface 1/3/4 in software is associated with physical port 4 in slot 3 on member 1.

On the 83xx Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. These are not a modular switches, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.



NOTE: If using breakout cables, the port designation changes to x:y, where x is the physical port and y is the lane when split to 4 x 10G or 4 x 25G. For example, the logical interface 1/1/4:2 in software is associated with lane 2 on physical port 4 in slot 1 on member 1.

On the 8400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/5 and 1/6.
 - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- *port*: Physical number of a port on a line module.

For example, the logical interface 1/1/4 in software is associated with physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
 - *member*: 1.
 - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
 - *member*: 1.
 - *tray*: 1 to 4.
 - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:
 - *member*: 1.
 - *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Overview

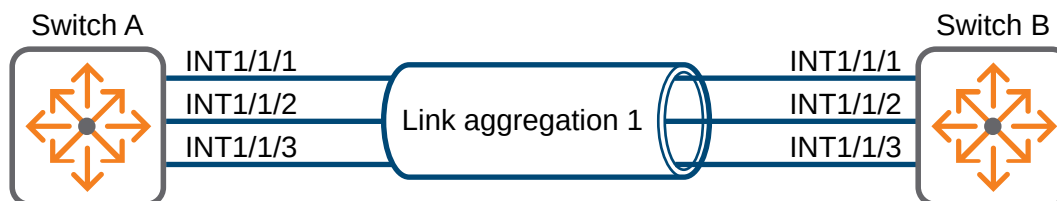
Ethernet link aggregation bundles multiple physical Ethernet links into one logical link, called a link aggregation group (LAG).

Link aggregation has the following benefits:

- Increased bandwidth beyond the limits of any single link. In an aggregate link, traffic is distributed across the member ports.
- Improved link reliability. The member ports dynamically back up one another. When a member port fails, its traffic is automatically switched to other member ports.

As shown in the following figure Device A and Device B are connected by three physical Ethernet links. These physical Ethernet links are combined into an aggregate link called link aggregation 1. The bandwidth of this aggregate link can reach up to the total bandwidth of the three physical Ethernet links. At the same time, the three Ethernet links back up one another. When a physical Ethernet link fails, the traffic originally intended for the failed link is switched to the remaining active links.

Figure 1: Ethernet link aggregation diagram



Aggregation group, member port, and aggregate interface

An aggregation group is a collection of physical interfaces that are bundled together for the purpose of load distribution and redundancy. These physical interfaces are called member ports. They are configured through a logical aggregate interface.

An aggregate interface can be one of the following types:

- **Layer 2:** The member ports of the corresponding Link Aggregation Group can only be Layer 2 Ethernet interfaces.
- **Layer 3:** The member ports of the corresponding Link Aggregation Group can only be Layer 3 interfaces.



NOTE: Layer 3 aggregation groups are not supported on the 6100 and 6200 Switch Series.

The effective port rate of an aggregate interface equals the total rate of its member ports. Only full duplex mode members are eligible for aggregation.

Link aggregation modes

An aggregation group operates in one of the following modes:

- **Static LAG:** In the static LAG mode of operation, Link failure is not detected as there is no keep alive PDU communication between the devices. A misconfiguration on one side can cause much trouble and be difficult to troubleshoot, because no signaling takes place between the two peers.
- **Dynamic LAG or LACP:** The local device and the peer device automatically maintain the aggregation states of the member ports, resulting in link failure being quickly detected by exchanging the PDU. LACP reduces the workload of network administrators.

Dynamic LAG uses LACP packets to establish the association between two peers. This configuration results in the reduction of the misconfiguration probability. Also, link failures are intelligently handled by two participating devices through the LACP protocol, which is adaptive/dynamic to these network failures.

Layer 2 aggregation groups and Layer 3 aggregation groups support both the static and dynamic modes.

LACP

Dynamic aggregation is implemented through the IEEE 802.3ad Link Aggregation Control Protocol (LACP).

LACP uses LACPDUs to exchange aggregation information between LACP-enabled devices. Each member port in a dynamic aggregation group can exchange information with its peer. When a member port receives an LACPDU, it compares the received information with information received on the other member ports. In this way, the two systems agree on which ports are placed in Selected state.

The LACPDU fields convey data for the LACP functions, including:

- System LACP priority
- System MAC address
- Port priority
- Port number
- Operational key

LACP operating modes

LACP can operate in active or passive mode.

- **Active mode:** When the LACP is operating in active mode on either end of a link, both ports can send PDUs. The "active" LACP initiates an LACP connection by sending LACPDUs. The "passive" LACP will wait for the remote end to initiate the link.
- **Passive mode:** When the LACP is operating in passive mode on a local member port and as its peer port, both ports cannot send PDUs.



NOTE: Two peer ports operating in "passive" mode will never establish an LACP link.

For an LACP LAG, one side must have LACP in active mode and the peer must have an LACP configuration of active or passive mode. If you do not enable LACP on a LAG, it is treated as a static LAG and the peer cannot negotiate LACP with the LAG.

LAG interface states

The output from the CLI commands `show lacp interfaces` and `show lacp interfaces multi-chassis` display the following interface states:

Interface state	Description
A - Active	An active LACP interface.
C - Collecting	Data frames are received through the aggregate link and sent onto the intended destination.
D - Distributing	Data frames are transmitted through the aggregate link to reach the intended destination.
F - Agregable	The link can be used as part of an aggregate.
E - Default neighbor state	The link has the default state of the neighbor switch.
I - Individual	The link is used as an individual link.
L - Long-timeout	With the long timeout, an LACPDU is sent every 30 seconds. If no response comes from its partner after three LACPDUs are sent (90 seconds), a timeout event occurs. The LACP state machine then transitions to the appropriate state based on its current state.
N - InSync	The physical port is connected to the aggregate port that was last chosen by the logical election. The state variable selected is still true.
O - OutofSync	The hardware might be out of sync with the modified protocol information. If the hardware also has a status of collecting, do not transmit frames because they will be misdelivered.
P - Passive	The port participates in the protocol, as long as it has an active partner.
S - Short-timeout	In the short timeout configuration, an LACPDU is sent every second. If no response comes from its partner after three LACPDUs are sent, a timeout event occurs. The LACP state machine then transitions to the appropriate state based on its current state.
X - State m/c expired	The "current while" timer has expired. The "current while" timer then restarts with the short-timeout enabled. The term <code>State m/c</code> refers to a state machine.

How static link aggregation groups are built

Reference port selection process

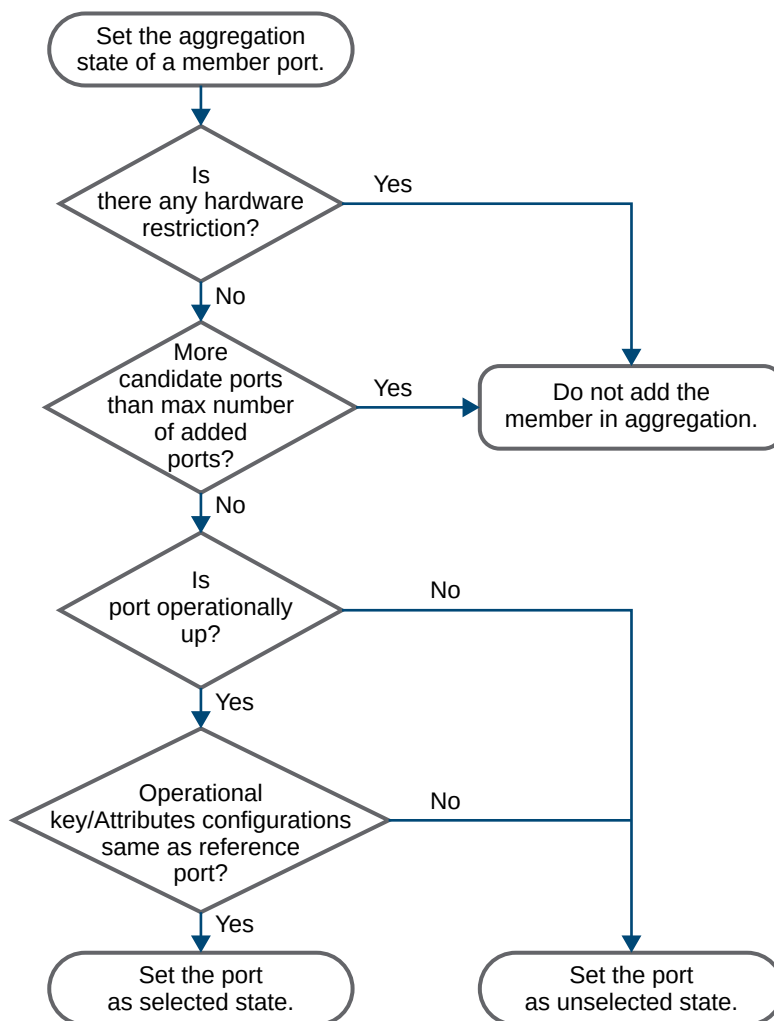
When setting the aggregation states of the ports in an aggregation group, the system automatically chooses a member port as the reference port. A selected port must have the same operational key and attribute configurations as the reference port.

The system chooses a reference port from the member ports in the up state. The first member interface which is operationally up is selected as reference port.

Setting the aggregation state of each member port

After the reference port is chosen, the system sets the aggregation state of each member port in the static aggregation group.

Figure 2: *Setting the aggregation state of a member port in a static aggregation group*



After the maximum limit of members is reached in a LAG, an additional port cannot be added to the aggregation group. If a port belongs to a card type with a different speed than the other aggregation members, the port can still be added to the aggregation group. If dynamic LAG is enabled, any port member with a speed different than other aggregation members is blocked or ineligible from the same aggregation group. Any operational keys/attributes or configuration changes might affect the aggregation states of the member ports.

How dynamic link aggregation groups are built

Choosing a reference port

The system chooses a reference port from the member ports in up state. A selected port must have the same operational key and attribute configurations as the reference port.

The process by which the local system (the actor) and the peer system (the partner) negotiate a reference port occurs as follows:

1. The two systems determine the system with the smaller system ID.
A system ID contains the system LACP priority and the system MAC address.
 - a. The two systems compare their LACP priority values.
The lower the LACP priority, the smaller the system ID. If the LACP priority values are the same, the two systems proceed to step b.
 - b. The two systems compare their MAC addresses.
The lower the MAC address, the smaller the system ID.
2. The system with the smaller system ID chooses the first operationally up port as the reference port.
A port ID contains a port priority and a port number. The lower the port priority, the smaller the port ID.

Setting the aggregation state of each member port

After the reference port is chosen, the system with the smaller system ID sets the state of each member port on its side.

The system with the greater system ID can detect the aggregation state changes on the peer system. The system with the greater system ID sets the aggregation state of local member ports the same as their peer ports.

When you aggregate interfaces in dynamic mode, follow these guidelines:

- A dynamic link aggregation group chooses only full-duplex ports as the selected ports.
- For stable aggregation and service continuity, do not change the operational key or attribute configurations on any member port.

LAG configuration guidelines

Aggregation member interface restrictions

- If any features in the following list are configured on the interface, you cannot assign an interface to a Layer 2 aggregation group:
 - MAC authentication
 - Port security
 - 802.1X
- Do not assign a reflector port for port mirroring to an aggregation group.

Requirements for adding interfaces

Keep in mind the following requirements when adding interfaces to a LAG:

- To determine the maximum number of LAG interfaces for your type of switch, look at the output from the `show capacities lag` command; however, the number of LAGs that can be created depends on the availability of the physical interface since each LAG interface needs at least one physical interface as a member link.

After the maximum limit of members is reached in a LAG, an additional port cannot be added to the aggregation group. If a port belongs to a card type with a different speed than the other aggregation members, the port can still be added to the aggregation group. If dynamic LAG is enabled, any port member with a speed different than other aggregation members is blocked or ineligible from the same aggregation group. Any operational keys/attributes or configuration changes might affect the aggregation states of the member ports.

- The nondefaults configuration on an interface is removed automatically when the interface is added to a link aggregation. For example: Assume that you remove a member interface from an existing LAG and add it to another LAG. The software removes the nondefault configurations on the interface when it is added to the new LAG.

Configuration consistency requirements

- Configure at least one active mode aggregation in two devices.
- For a successful static aggregation, make sure the ports at both ends of each link are in the same aggregation state.
- For a successful dynamic aggregation, make sure the peer ports of the ports aggregated at one end are also aggregated, and that one of the ends is configured as "active". The two ends can automatically negotiate the aggregation state of each member port.

Removing interfaces

- Deleting an aggregate interface also deletes its aggregation group and causes all member ports to leave the aggregation group.
- When a member interface is removed from a LAG:
 - **6100, 6200, 6300, and 6400 switches:** The interface goes to its default status of `unshut`.
 - **8320, 8325, 8360, or 8400 switches:** The interface becomes disabled.

Disabling an interface

When an interface LAG is disabled with the `shutdown` command, all its members also become operationally down.

Layer 2 aggregation groups

All switches support static and dynamic layer 2 aggregation groups.



NOTE: On the 6400 Switch Series, port identification differs. Line card ports start at 1/3/1.

Configuring a Layer 2 static aggregation group

Prerequisites

You must be in the global configuration context: `switch(config)#`

Procedure

1. Create a Layer 2 aggregate interface and access the Layer 2 aggregate interface view by entering:

```
switch(config)# interface lag <ID>
```

The range of the LAG interface ID is 1 to 256.

While creating the Layer 2 aggregate interface, the system automatically creates a Layer 2 static aggregation group numbered the same.

2. Set the operational state of every interface in the LAG to up by entering:

```
switch(config-lag-if)# no shutdown
```



NOTE: This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:

```
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
```

3. On the 8320, 8325, 8360, and 8400, disable routing by entering:

```
switch(config-lag-if)# no routing
```

See the *Command-Line Interface Guide* for your switch and software version for more information about the `no routing` command.



NOTE: On the 6100 and 6200 Switch Series, routing is not supported on physical interfaces. On the 6300 and 6400 Switch Series, routing is disabled by default.

4. Assign a native VLAN ID to a trunk interface on the LAG by entering:

```
switch(config-lag-if)# vlan trunk native <VLAN-ID>
```

For example:

```
switch(config-lag-if)# vlan trunk native 1
```

5. Use the following steps to add a maximum of 16 interfaces to the LAG:

- a. To assign an interface to the LAG:

```
switch(config-lag-if)# interface <PORT-ID>
```

To assign a range of interfaces to a LAG:

```
switch(config-lag-if)# interface <PORT-ID>--<PORT-ID>
```

For example:

```
switch(config-lag-if) # interface 1/1/1-1/1/4
```

See the *Command-Line Interface Guide* for your switch and software version for more information about the interface <PORT-ID> command.

b. Assign an ID to the LAG:

```
switch(config-if) # lag <ID>
```

For example:

```
switch(config-if-<1/1/1-1/1/4>) # lag 100
```

c. Set the administrative state of the member interface to up:

```
switch(config-if-<1/1/1-1/1/4>) # no shutdown
```

6. View the configuration by entering the following:

For 6100, 6200, 6300, and 6400 switch series:

```
switch(config-if-<1/1/1-1/1/4>) # show running-config
```

Current configuration:

```
!  
vlan 1  
interface lag 100  
    no shutdown  
    vlan trunk native 1  
    vlan trunk allowed all  
interface 1/1/1  
    no shutdown  
    lag 100  
interface 1/1/2  
    no shutdown  
    lag 100  
interface 1/1/3  
    no shutdown  
    lag 100  
interface 1/1/4  
    no shutdown  
    lag 100
```

```
switch(config-if-<1/1/1-1/1/4>) # show lacp aggregates
```

```
Aggregate name      : lag100  
Interfaces          : 1/1/3 1/1/1 1/1/4 1/1/2  
Heartbeat rate     : N/A  
Hash                : l3-src-dst  
Aggregate mode      : Off
```

For 8320, 8325, 8360, and 8400 switch series:

```
switch(config-if-<1/1/1-1/1/4>) # show running-config
```

Current configuration:

```
!  
vlan 1  
interface lag 100  
    no shutdown  
    no routing  
    vlan trunk native 1  
    vlan trunk allowed all  
interface 1/1/1
```



```

    no shutdown
    lag 100
interface 1/1/2
    no shutdown
    lag 100
interface 1/1/3
    no shutdown
    lag 100
interface 1/1/4
    no shutdown
    lag 100

switch(config-if-<1/1/1-1/1/4>) # show lacp aggregates

Aggregate name      : lag100
Interfaces          : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate      : N/A
Hash                : 13-src-dst
Aggregate mode      : Off

```

Configuring a Layer 2 dynamic aggregation group

Prerequisites

You must be in the global configuration context: `switch(config) #`

Procedure

1. Create a Layer 2 aggregate interface and access the Layer 2 aggregate interface view by entering:

```
switch(config) # interface lag <ID>
```

The range of the LAG interface ID is 1 to 256.

While creating the Layer 2 aggregate interface, the system automatically creates a Layer 2 dynamic aggregation group numbered the same.

2. Set the operational state of every interface in the LAG to up by entering:

```
switch(config-lag-if) # no shutdown
```



NOTE: This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:

```
switch(config) # interface 1/1/2
switch(config-if) # no shutdown
```

3. On the 8320, 8325, 8360, and 8400, disable routing by entering:

```
switch(config-lag-if) # no routing
```

See the *Command-Line Interface Guide* for your switch and software version for more information about the `no routing` command.



NOTE: On the 6100 and 6200 Switch Series, routing is not supported on physical interfaces. On the 6300 and 6400 Switch Series, routing is disabled by default.

4. Configure the aggregation group to operate in dynamic mode by entering:

```
switch(config-lag-if) # lacp mode {active | passive}
```

For example:

```
switch(config-lag-if) # lacp mode active
```

5. Configure the aggregation group to operate in fast or slow mode by entering:

```
switch(config-lag-if) # lacp rate {fast | slow}
```

For example:

```
switch(config-lag-if) # lacp rate fast
```

6. Assign a native VLAN ID to a trunk interface by entering:

```
switch(config-lag-if) # vlan trunk native <VLAN-ID>
```

For example:

```
switch(config-lag-if) # vlan trunk native 1
```

7. Use the following steps to add a maximum of 16 interfaces to the LAG:

- a. To assign an interface to the LAG:

```
switch(config-lag-if) # interface <PORT-ID>
```

To assign a range of interfaces to a LAG:

```
switch(config-lag-if) # interface <PORT-ID>--<PORT-ID>
```

For example:

```
switch(config-lag-if) # interface 1/1/1-1/1/4
```

See the *Command-Line Interface Guide* for your switch and software version for more information about the `interface <PORT-ID>` command.

- b. Assign an ID to the LAG:

```
switch(config-if) # lag <ID>
```

For example:

```
switch(config-if-<1/1/1-1/1/4>) # lag 20
```

- c. Set the administrative state of the member interface to up:

```
switch(config-if-<1/1/1-1/1/4>) # no shutdown
```

8. View the configuration by entering:

For 6100, 6200, 6300, and 6400 switch series:

```
switch(config-if-<1/1/1-1/1/4>) # show running-config
```

Current configuration:

```
!  
vlan 1  
interface lag 20  
    no shutdown  
    vlan trunk native 1
```

```

    vlan trunk allowed all
    lacp mode active
    lacp rate fast
interface 1/1/1
    no shutdown
    lag 20

switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates

Aggregate name      : lag100
Interfaces          : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate      : Fast
Hash                : 13-src-dst
Aggregate mode      : Active

```

For 8320, 8325, 8360, and 8400 switch series:

```

switch(config-if-<1/1/1-1/1/4>)# show running-config

Current configuration:
!
vlan 1
interface lag 20
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed all
    lacp mode active
    lacp rate fast
interface 1/1/1
    no shutdown
    lag 20

switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates

Aggregate name      : lag100
Interfaces          : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate      : Fast
Hash                : 13-src-dst
Aggregate mode      : Active

```

Layer 3 aggregation groups

Layer 3 aggregation groups are supported on all switch series except 6100 and 6200 Switch Series.

Configuring a Layer 3 static aggregation group

Prerequisites

You must be in the global configuration context: `switch(config)#`

Procedure

1. Create a Layer 3 aggregate interface and access the Layer 3 aggregate interface view by entering:

```
switch(config)# interface lag <ID>
```

The range of the LAG interface ID is 1 to 256.

While creating the Layer 3 aggregate interface, the system automatically creates a Layer 3 static aggregation group numbered the same.

2. Set the operational state of every interface in the LAG to up by entering:

- For 6300 and 6400 switch series:

```
switch(config-lag-if) # no shutdown
switch(config-lag-if) # routing
```



NOTE: This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:

```
switch(config) # interface 1/1/2
switch(config-if) # no shutdown
switch(config-if) # routing
```

- For 8320, 8325, 8360, and 8400 switch series:

```
switch(config-lag-if) # no shutdown
```



NOTE: This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:

```
switch(config) # interface 1/1/2
switch(config-if) # no shutdown
```

3. Set the IP address on the LAG interface by entering:

```
switch(config-lag-if) # ip address <IPV4-ADDR>/<MASK>
```

For example:

```
switch(config-lag-if) # ip address 192.0.2.1/30
```

4. Use the following steps to add a maximum of 16 interfaces to the LAG:

- a. To assign an interface to the LAG:

```
switch(config-lag-if) # interface <PORT-ID>
```

To assign a range of interfaces to a LAG:

```
switch(config-lag-if) # interface <PORT-ID>--<PORT-ID>
```

For example:

```
switch(config-lag-if) # interface 1/1/1-1/1/4
```

See the *Command-Line Interface Guide* for your switch and software version for more information about the `interface <PORT-ID>` command.

- b. Assign an ID to the LAG:

```
switch(config-if) # lag <ID>
```

For example:

```
switch(config-if-<1/1/1-1/1/4>)# lag 100
```

- c. Set the administrative state of the member interface to up:

```
switch(config-if-<1/1/1-1/1/4>)# no shutdown
```

5. View the configuration by entering the following:

For 6300 and 6400 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
```

Current configuration:

```
!  
vlan 1  
interface lag 100  
    no shutdown  
    routing  
    ip address 192.0.2.1/30  
interface 1/1/1  
    no shutdown  
    lag 100  
interface 1/1/2  
    no shutdown  
    lag 100  
interface 1/1/3  
    no shutdown  
    lag 100  
interface 1/1/4  
    no shutdown  
    lag 100
```

```
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
```

```
Aggregate name      : lag100  
Interfaces          : 1/1/3 1/1/1 1/1/4 1/1/2  
Heartbeat rate      : N/A  
Hash                : 13-src-dst  
Aggregate mode      : Off
```

For 8320, 8325, 8360, and 8400 switch series:

```
switch(config-if-<1/1/1-1/1/4>)# show running-config
```

Current configuration:

```
!  
vlan 1  
interface lag 100  
    no shutdown  
    ip address 192.0.2.1/30  
interface 1/1/1  
    no shutdown  
    lag 100  
interface 1/1/2  
    no shutdown  
    lag 100  
interface 1/1/3  
    no shutdown  
    lag 100  
interface 1/1/4  
    no shutdown  
    lag 100
```

```
switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates
```

```
Aggregate name      : lag100
Interfaces          : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate     : N/A
Hash                : l3-src-dst
Aggregate mode      : Off
```

Configuring a Layer 3 dynamic aggregation group

Prerequisites

You must be in the global configuration context: `switch(config)#`

Procedure

1. Create a Layer 3 aggregate interface and access the Layer 3 aggregate interface view by entering:

```
switch(config)# interface lag <ID>
```

The range of the LAG interface ID is 1 to 256.

While creating the Layer 3 aggregate interface, the system automatically creates a Layer 3 dynamic aggregation group numbered the same.

2. Set the operational state of every interface in the LAG to up by entering:

- For 6300 and 6400 switch series:

```
switch(config-lag-if) # no shutdown
switch(config-lag-if) # routing
```



NOTE: This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:

```
switch(config)# interface 1/1/2
switch(config-if) # no shutdown
switch(config-if) # routing
```

- For 8320, 8325, 8360, and 8400 switch series:

```
switch(config-lag-if) # no shutdown
```



NOTE: This command does not impact the administrative state of the member interfaces because the command was entered at the level of the LAG. To change the administrative state of a member interface, enter the command at the interface level. For example:

```
switch(config)# interface 1/1/2
switch(config-if) # no shutdown
```

3. Configure the aggregation group to operate in dynamic mode by entering:

```
switch(config-lag-if) # lacp mode {active | passive}
```

For example:

```
switch(config-lag-if) # lacp mode active
```

4. Configure the aggregation group to operate in fast or slow mode by entering:

```
switch(config-lag-if) # lacp rate {fast | slow}
```

For example:

```
switch(config-lag-if) # lacp rate fast
```

5. Set the IP address on the LAG interface by entering:

```
switch(config-lag-if) # ip address <IPV4-ADDR>/<MASK>
```

For example:

```
switch(config-lag-if) # ip address 192.0.3.1/30
```

6. Use the following steps to add a maximum of 16 interfaces to the LAG:

- a. To assign an interface to the LAG:

```
switch(config-lag-if) # interface <PORT-ID>
```

To assign a range of interfaces to a LAG:

```
switch(config-lag-if) # interface <PORT-ID>--<PORT-ID>
```

For example:

```
switch(config-lag-if) # interface 1/1/1-1/1/4
```

See the *Command-Line Interface Guide* for your switch and software version for more information about the interface <PORT-ID> command.

- b. Assign an ID to the LAG:

```
switch(config-if) # lag <ID>
```

For example:

```
switch(-<1/1/1-1/1/4>) # lag 100
```

- c. Set the administrative state of the member interface to up:

```
switch(-<1/1/1-1/1/4>) # no shutdown
```

7. View the configuration by entering:

For 6300 and 6400 switch series:

```
switch(config-if-<1/1/1-1/1/4>) # show running-config
```

Current configuration:

```
!  
vlan 1  
interface lag 100  
    no shutdown  
    routing  
    ip address 192.0.3.1/30  
    lacp mode active  
    lacp rate fast  
interface 1/1/1  
    no shutdown  
    lag 100  
interface 1/1/2  
    no shutdown
```

```

    lag 100
interface 1/1/3
    no shutdown
    lag 100
interface 1/1/4
    no shutdown
    lag 100

switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates

Aggregate name      : lag100
Interfaces          : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate      : Fast
Hash                : 13-src-dst
Aggregate mode      : Active

```

For 8320, 8325, 8360, and 8400 switch series:

```

switch(config-if-<1/1/1-1/1/4>)# show running-config

Current configuration:
!
vlan 1
interface lag 100
    no shutdown
    ip address 192.0.3.1/30
    lacp mode active
    lacp rate fast
interface 1/1/1
    no shutdown
    lag 100
interface 1/1/2
    no shutdown
    lag 100
interface 1/1/3
    no shutdown
    lag 100
interface 1/1/4
    no shutdown
    lag 100

switch(config-if-<1/1/1-1/1/4>)# show lacp aggregates

Aggregate name      : lag100
Interfaces          : 1/1/3 1/1/1 1/1/4 1/1/2
Heartbeat rate      : Fast
Hash                : 13-src-dst
Aggregate mode      : Active

```

Removing a LAG

Prerequisites

You must be in the global configuration context: `switch(config)#`

Procedure

1. Delete the LAG. Enter:

```
switch(config)# no interface lag <ID>
```


For example:

```
switch(config)# no interface lag 100
```

All interfaces assigned to the LAG are automatically removed from the LAG as part of the deletion process of the LAG. After removing a physical interface from a LAG,

- **6100, 6200, 6300, and 6400 switches:** The interface associated with the LAG becomes layer 2 ports with the default layer 2 configurations and `admin` status enabled.
- **8320, 8235, 8360, and 8400 switches:** The interface associated with the LAG becomes layer 3 ports with default layer 3 configurations and administrative down.

Removing an interface from a LAG

Prerequisites

You must be in the global configuration context: `switch(config)#`

Procedure

1. Remove an interface from the LAG. Enter:

```
switch(config)# interface <PORT-NUM>
switch(config-if)# no lag <ID>
```

For example:

```
switch(config)# interface 1/1/1
switch(config-if)# no lag 100
switch(config-if)# show running-config
...
!
vlan 1
interface lag 100
interface 1/1/1
interface 1/1/2
    lag 100
switch(config-if)#
```

To assign a range of interfaces to a LAG:

```
switch(config-lag-if)# interface <PORT-ID>-<PORT-ID>
```

For example:

```
switch(config-lag-if)# interface 1/1/1-1/1/4
```

After removing a physical interface from a LAG:

- **6100, 6200, 6300, and 6400 switches:** The interface associated with LAG becomes layer 2 ports with default layer 2 configurations and with admin status of `enabled`.
- **8320, 8235, 8360, and 8400 switches:** The interface associated with the LAG becomes L3 ports with default L3 configurations and administrative down. For example, suppose interface 1/1/1 was part of LAG 3 and you had administratively enabled the interface. If you later remove interface 1/1/1 from

LAG 3, the administrative status automatically changes to down. If you want to use the interface again, you must administratively enable it again.

Changing the LAG membership for an interface

Prerequisites

You must be in the global configuration context: `switch(config)#`

Procedure

1. Remove an interface from the LAG. Enter:

```
switch(config)# interface <PORT-NUM>
switch(config-if)# no lag <ID>
```

For example:

For 6100, 6200, 6300, and 6400 switch series:

```
switch(config)# interface 1/1/1
switch(config-if)# no lag 100
switch(config-if)# show running-config
Current configuration:
!
...
!
vlan 1
interface lag 100
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
interface 1/1/2
    no shutdown
    lag 100
switch(config-if)#
```

For 8320, 8325, 8360, and 8400 switch series:

```
switch(config)# interface 1/1/1
switch(config-if)# no lag 100
switch(config-if)# show running-config
Current configuration:
!
...
!
vlan 1
interface lag 100
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
interface 1/1/2
    no shutdown
    lag 100
switch(config-if)#
```

After removing a physical interface from a LAG, the interface associated with the LAG becomes L3 ports with default L3 configurations and administrative down. For example, suppose interface 1/1/1 was part of LAG 3 and you had administratively enabled the interface. If you later remove interface 1/1/1 from LAG 3, the administrative status automatically changes to down. If you want to use the interface again, you must administratively enable it again.

On 6100 and 6200 Switch Series, after removing a physical interface from a LAG, the interface associated with the LAG becomes layer 2 ports with default layer 2 configurations and `admin` status enabled.

2. Create the LAG to which you want to add the interface:

```
switch(config-if) # interface lag 10
```

For example:

For 6100, 6200, 6300, and 6400 switch series:

```
switch(config-if) # interface lag 10  
switch(config-lag-if) # no shutdown  
switch(config-lag-if) # vlan trunk native 1
```

For 8320, 8325, 8360, and 8400 switch series:

```
switch(config-if) # interface lag 10  
switch(config-lag-if) # no shutdown  
switch(config-lag-if) # no routing  
switch(config-lag-if) # vlan trunk native 1
```

3. Add the interface from Step 1 to the newly created LAG:

```
switch(config) # interface 1/1/1  
switch(config-if) # lag 10
```

For example:

For 6100, 6200, 6300, and 6400 switch series:

```
switch(config) # interface 1/1/1  
switch(config-if) # lag 10  
switch(config-if) # show running-config  
Current configuration:  
!  
...  
!  
vlan 1  
interface lag 10  
    no shutdown  
    vlan trunk native 1  
    vlan trunk allowed all  
interface lag 100  
    no shutdown  
    vlan trunk native 1  
    vlan trunk allowed all  
interface 1/1/1  
    lag 10  
interface 1/1/2  
    no shutdown  
    lag 100
```

For 8320, 8325, 8360, and 8400 switch series:

```
switch(config) # interface 1/1/1  
switch(config-if) # lag 10  
switch(config-if) # show running-config
```

```

Current configuration:
!
...
!
vlan 1
interface lag 10
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed all
interface lag 100
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
    lag 10
interface 1/1/2
    no shutdown
    lag 100

```

Notice that interface 1/1/1 in the previous example is still not active, even though it has been added to LAG 10. To change the administrative state of the member interface, enter the `no shutdown` command at the interface level.

For example:

For 6100, 6200, 6300, and 6400 switch series:

```

switch(config-if)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# show running-config
Current configuration:
!
...
!
vlan 1
interface lag 10
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
interface lag 100
    no shutdown
    vlan trunk native 1
    vlan trunk allowed all
interface 1/1/1
    no shutdown
    lag 10
interface 1/1/2
    no shutdown
    lag 100

```

For 8320, 8325, 8360, and 8400 switch series:

```

switch(config-if)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# show running-config
Current configuration:
!
...
!
vlan 1
interface lag 10
    no shutdown

```

```

no routing
vlan trunk native 1
vlan trunk allowed all
interface lag 100
no shutdown
no routing
vlan trunk native 1
vlan trunk allowed all
interface 1/1/1
no shutdown
lag 10
interface 1/1/2
no shutdown
lag 100

```

Configuration of an aggregate interface

Configuring the description of an aggregate interface

You can configure the description of an aggregate interface for administration purposes, for example, describing the purpose of the interface.

Prerequisites

You must be in the global configuration context: `switch(config)#`

Procedure

1. Create a Layer 3 aggregate interface and enter Layer 3 aggregate interface view by entering:

```
switch(config)# interface lag <ID>
```

2. Configure the description of the aggregate interface:

```
switch(config-if)# description <text>
```

Setting the MTU for a Layer 2 member link interface

Prerequisites

You must be in the global configuration context: `switch(config)#`

Procedure

1. Enter a Layer 2 member link interface view by entering:

```
switch(config)# interface <INTERFACE-ID>
```

2. Set the MTU for the Layer 2 member link interface:

```
switch(config-if)# mtu <VALUE>
```

See the *Command-Line Interface Guide* for your switch and software version for more information about the `mtu <VALUE>` command. When allowing jumbo frames under a layer 2 aggregation interface, make sure that the MTU value is set appropriately under all member interfaces.

Setting the MTU for a Layer 3 aggregate interface



NOTE: Layer 3 aggregation groups are not supported on the 6100 and 6200 Switch Series.

The MTU of an interface affects IP packets fragmentation and reassembly on the interface.

Prerequisites

You must be in the global configuration context: `switch(config)#`

Procedure

1. Enter Layer 3 aggregate interface view by entering:

```
switch(config)# interface lag <INTERFACE-ID>
```

2. Set the MTU for the Layer 3 aggregate interface:

```
switch(config-lag-if)# ip mtu <VALUE>
```

See the *Command-Line Interface Guide* for your switch and software version for more information about the `ip mtu <VALUE>` command. When allowing jumbo frames under a layer 2 aggregation interface, make sure that the MTU value is set appropriately under all member interfaces.



NOTE: If the IP MTU is configured as 9198, the MTU on the physical interfaces must also be configured as 9198.

Impact of shutting down or bringing up an aggregate interface

By default, an aggregate interface is down. Shutting down or bringing up an aggregate interface affects the aggregation states and link states of member ports in the corresponding aggregation group as follows:

- When an aggregate interface is shut down, all Selected ports in the corresponding aggregation group become Unselected ports and all member ports go to an operationally down state.
- When an aggregate interface is brought up, the aggregation states of member ports in the corresponding aggregation group are recalculated. LAG members, which are administratively up, will become operationally up. The members that are not administratively up will be in the same state and not made eligible for aggregation.

Shutting down an aggregate interface

Prerequisites

You must be in the global configuration context: `switch(config)#`

Procedure

1. Enter Layer 3 aggregate interface view by entering:

```
switch(config)# interface lag <ID>
```

2. Shut down the aggregate interface:

```
switch(config-lag-if) # shutdown
```

Supported hashing algorithms

- Source MAC and destination MAC
- Source IP and destination IP
- Source port and destination port.

LACP configuration settings

Task	Command	Example
Setting the LACP mode to active or passive.	lacp mode {active passive}	switch(config-lag-if) # lacp mode active
Setting the LACP mode to off.	no lacp mode {active passive}	switch(config-lag-if) # no lacp mode active
Setting the hash type.	For 6100 and 8400 Switch Series: lacp hash [12-src-dst 13-src-dst 14-src-dst] For 8320, 8325, 6200, 6300, and 6400 Switch Series: hash [12-src-dst 13-src-dst]	For 6100 and 8400 Switch Series: switch(config) # lacp hash 12-src-dst For 8320, 8325, 6200, 6300, and 6400 Switch Series: switch(config-lag-if) # hash 12-src-dst
Setting the LACP rate to fast.	lacp rate fast	switch(config) # interface lag 1 switch(config-lag-if) # lacp rate fast
Setting the LACP rate to slow.	lacp rate slow	switch(config) # interface lag 1 switch(config-lag-if) # lacp rate slow

Table Continued

Task	Command	Example
Applying shutdown on the LAG port.	shutdown	switch(config)# interface lag 1 switch(config-lag-if)# shutdown
Resetting every interface in the LAG to the default (up)	no shutdown	switch(config-lag-if)# no shutdown

Interface LACP settings

Task	Command	Example
Setting the LACP port ID.	lacp port-id <ID>	switch(config-if)# lacp port-id 100
Setting the LACP port ID to the default.	no lacp port-id	switch(config-if)# no lacp port-id
Setting the LACP port priority.	lacp port-priority <PORT-PRIORITY>	switch(config-if)# lacp port-priority 100
Setting the LACP port priority to the default	no lacp port-priority	switch(config-if)# no lacp port-priority

Configuration verification

Task	Command	Example
Viewing LACP global information	<u><code>show lacp configuration</code></u>	<pre>switch# show lacp configuration System-id : 70:72:cf:ef:fc:d9 System-priority : 65534 Hash : 13-src-dst</pre> The output displayed for the <code>show lacp configuration</code> is from the 8400 series switch.
Viewing LACP aggregate information	<u><code>show lacp aggregates</code></u>	<pre>switch# show lacp aggregates Aggregate-name : lag100 Aggregated-interfaces : 1/1/2 Heartbeat rate : N/A Hash : 13-src-dst Aggregate mode : off Aggregate-name : lag110 Aggregated-interfaces : 1/1/1, 1/1/3 Heartbeat rate : slow Hash : 13-src-dst Aggregate mode : active</pre>
Viewing LACP aggregate information for a LAG	<u><code>show lacp aggregates lag100</code></u>	<pre>switch# show lacp aggregates lag100 Aggregate-name : lag100 Aggregated-interfaces : Heartbeat rate : N/A Hash : 13-src-dst Aggregate mode : off</pre>
Viewing LACP interface details	<u><code>show lacp interfaces</code></u>	<pre>switch# show lacp interfaces</pre> The output is too wide to display in a column. The command output is provided in the CLI topic for the command.

BFD reports a LAG as down even when healthy links are still available

Symptom



NOTE: BFD is not supported on the 6100 and 6200 Switch Series.

The Bidirectional Forward Detection (BFD) feature reports a Link Aggregation (LAG), as being down, even though there are healthy LAG links available. The LAG, containing the downed link, will eventually rebalance the traffic to its other links.

Cause

This notification occurs when the minimum BFD control packet reception interval is set at a faster rate than the Link Aggregation Control Protocol (LACP) rate and LAG rebalancing occurs. BFD assumes that the link is down without realizing that the LAG is rebalancing the traffic load.

Action

1. Set the minimum BFD control packet reception interval to a slower rate than the LACP rate or set the LACP rate to a faster rate than the minimum BFD control packet reception interval.
 - a. To find the current settings of the minimum BFD control packet reception interval, enter the `show running-config` command.

The minimum BFD control packet reception interval setting is listed as `bfd min-receive-interval` in the command output and the measurement is in ms.
 - b. To find the current rate of LACP, enter the `show lacp aggregates` command.

The LACP rate is listed as the `Heartbeat rate` in the command output.
 - c. To change the minimum BFD control packet reception interval, enter the `bfd min-receive-interval` command.
 - d. To change the LACP rate, enter the `lacp rate {fast | slow}` command.

LACP and LAG commands

description

Syntax

```
description <TEXT>
```

```
no description <TEXT>
```

Description

Provides a brief description of the LAG interface. The description text is saved in the configuration of the LAG. It is available even after a reboot.

The `no` form of this command removes the description of the LAG interface from the configuration.

Command context

```
config-lag-if
```

Parameters

<TEXT>

Specifies the description of the LAG interface.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# interface lag 10
switch(config-lag-if)# description This LAG is used for an example.
switch(config-lag-if)# show running-config
...
vlan 1
interface lag 10
    description This LAG is used for an example.
```

```
interface lag 60
switch(config-lag-if) #
```

hash

Syntax

```
hash [l2-src-dst | l3-src-dst | l4-src-dst]
```

Description

This command controls the selection of an interface in a group of aggregate interfaces. The hash type value helps transmit a frame. This configuration must be done at the LAG interface level.

Command context

```
config-lag-if
```

Parameters

l2-src-dst

Specifies the load-balancing calculation to include only Layer 2 items, such as source and destination MAC addresses.

l3-src-dst

Specifies the load-balancing calculation to include only Layer 3 items, such as source and destination IP addresses. Default setting.

l4-src-dst

Specifies the load-balancing calculation to include only Layer 4 items, such as source and destination UDP/TCP ports.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config-lag-if) # hash l2-src-dst
```

interface lag

Syntax

```
interface lag <ID>
no interface lag <ID>
```

Description

Creates a Link Aggregation Group (LAG) interface represented by an ID.

The `no` form of this command deletes a LAG interface represented by an ID.

Command context

```
config
```

Parameters

<ID>

Specifies a LAG interface ID.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Keep in mind the following requirements when adding interfaces to a LAG:

- To determine the maximum number of LAG interfaces for your type of switch, look at the output from the `show capacities lag` command; however, the number of LAGs that can be created depends on the availability of the physical interface since each LAG interface needs at least one physical interface as a member link.
- After the maximum limit of members is reached in a LAG, an additional port cannot be added to the aggregation group. If a port belongs to a card type with a different speed than the other aggregation members, the port can still be added to the aggregation group. If dynamic LAG is enabled, any port member with a speed different than other aggregation members is blocked or ineligible from the same aggregation group. Any operational keys/attributes or configuration changes might affect the aggregation states of the member ports.
- The nondefaults configuration on an interface is removed automatically when the interface is added to a link aggregation. For example: Assume that you remove a member interface from an existing LAG and add it to another LAG. The software removes the nondefault configurations on the interface when it is added to the new LAG.

Examples

Creating a Link Aggregation Group (LAG) interface represented by an ID of 100:

```
switch(config)# interface lag 100
```

Deleting a Link Aggregation Group (LAG) interface represented by an ID of 100:

```
switch(config)# no interface lag 100
```

ip address

Not supported on the 6100 and 6200 Switch Series.

Syntax

```
ip address <IPV4-ADDR>/<MASK> [secondary]
```

```
no ip address <IPV4-ADDR>/<MASK> [secondary]
```

Description

Sets an IPv4 address and subnet mask to a LAG interface. One primary and up to 31 secondary address can be configured per interface.

The `no` form of this command removes the IPv4 address from the interface.

Command context

```
config-lag-if
```

Parameters

<IPV4-ADDR>

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 32.

secondary

Specifies a secondary IP address.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting an IP address on the LAG interface 1 to 198.51.100.1 with a mask of 24 bits:

```
switch(config)# interface lag 1
switch(config-lag-if)# ip address 198.51.100.1/24
```

Removing the IP address 198.51.100.1 with a mask of 24 bits from LAG interface 1:

```
switch(config)# interface lag 1
switch(config-lag-if)# no ip address 198.51.100.1/24
```

ipv6 address

Not supported on the 6100 and 6200 Switch Series.

Syntax

```
ipv6 address <IPV6-ADDR>/<MASK>
```

```
no ipv6 address <IPV6-ADDR>/<MASK>
```

Description

Sets an IPv6 address and subnet mask to a LAG interface.

The `no` form of this command removes the IPv6 address from the interface.

Command context

```
config-lag-if
```

Parameters

<IPV6-ADDR>

Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a quartet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

<MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the IPv6 address on LAG interface 1 to 2001:0db8:85a3::8a2e:0370:7334 with a mask of 24 bits:

```
switch(config)# interface lag 1
switch(config-lag-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Removing the IP address 2001:0db8:85a3::8a2e:0370:7334 with mask of 24 bits with a mask of 24 bits from LAG interface 1:

```
switch(config)# interface lag 1
switch(config-lag-if)# no ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

lacp hash

Supported only on the 6100 and 8400 Switch Series.

Syntax

```
lacp hash [l2-src-dst | l3-src-dst | l4-src-dst]
```

Description

This command controls the selection of an interface in a group of aggregate interfaces. The hash type value helps transmit a frame. This configuration must be done at the global level.

Command context

```
config
```

Parameters

l2-src-dst

Specifies the load-balancing calculation to include only Layer 2 items, such as source and destination MAC addresses.

l3-src-dst

Specifies the load-balancing calculation to include only Layer 3 items, such as source and destination IP addresses.

l4-src-dst

Specifies the load-balancing calculation to include only Layer 4 items, such as source and destination UDP/TCP ports.

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config)# lacp hash l2-src-dst
```

lacp mode

Syntax

```
lacp mode {active | passive}
no lacp mode {active | passive}
```

Description

Sets an LACP mode to active or passive.

The `no` form of this command sets the LACP mode to `off`, returning the LAG to a static mode aggregation.

Command context

`config-lag-if`

Parameters

active

Specifies that the local switch will transmit LACP Data Units (LACPDUs) to attempt to negotiate with the remote device.

passive

Specifies that the local switch will listen for LACPDUs from the remote device for LACP negotiation.



NOTE: A momentary traffic drop occurs because LACP partners reconverge when changing the mode from active to passive or passive to active.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the LACP mode to `active`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp mode active
```

Setting the LACP mode to `off`:

```
switch(config)# interface lag 1
switch(config-lag-if)# no lacp mode active
```

lacp port-id

Syntax

`lacp port-id <PORT-ID>`

`no lacp port-id`

Description

Sets the LACP port ID value of the member interface of the LAG.

The `no` form of this command removes the LACP port ID value from the interface.

Command context

`config-if`

Parameters

`<PORT-ID>`

Specifies a port ID value. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting an LACP port ID to a value of 10:

```
switch(config-if)# lacp port-id 10
```

Removing the LACP port ID value:

```
switch(config-if)# no lacp port-id
```

lacp port-priority

Syntax

```
lacp port-priority <PORT-PRIORITY>
```

```
no lacp port-priority
```

Description

Sets an LACP port priority value for the member interface of the LAG.

The `no` form of this command reverts the LACP port priority to the default, which is 1.

Command context

```
config-if
```

Parameters

<PORT-PRIORITY>

Specifies a port priority value. Range: 1 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting a LACP port priority value of 10:

```
switch(config-if)# lacp port-priority 10
```

Reverting the LACP port ID to the default:

```
switch(config-if)# no lacp port-priority
```

lacp rate

Syntax

```
lacp rate {fast | slow}
```

```
no lacp rate
```

Description

Sets an LACP heartbeat request time to fast or slow.

The `no` form of the command sets an LACP rate to `slow`.

Command context

`config-lag-if`

Parameters

fast

Specifies the heartbeat request to every second, and the timeout period is a three-consecutive heartbeat loss that is 3 seconds.

slow

Specifies the heartbeat request to every 30 seconds. The timeout period is three-consecutive heartbeat loss that is 90 seconds. Default setting.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the LACP heartbeat request time to `fast`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp rate fast
```

Resetting the LACP heartbeat request time to the default, which is `slow`:

```
switch(config)# interface lag 1
switch(config-lag-if)# no lacp rate
```

Another way to set the LACP heartbeat request time to the default, which is `slow`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp rate slow
```

lacp system-priority

Syntax

```
lacp system-priority <SYSTEM-PRIORITY-VALUE>
no lacp system-priority
```

Description

Sets a Link Aggregation Control Protocol (LACP) system priority.

The `no` form of this command sets an LACP system priority to the default, which is 65534.

Command context

`config`

Parameters

<SYSTEM-PRIORITY-VALUE>

Specifies a system priority value. Range: 0 to 65535.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting a Link Aggregation Control Protocol (LACP) system priority to 100:

```
switch(config)# lacp system-priority 100
```

Setting an LACP system priority to the default (65534):

```
switch(config)# no lacp system-priority
```

A momentary traffic drop can be seen in case the LACP state machine must renegotiate.

lag

Syntax

```
lag <ID>
```

```
no lag <ID>
```

Description

Adds an interface to a specified LAG interface ID.

The `no` form of this command removes an interface from a specified LAG interface ID. The member loses its LACP configuration when removed from the LAG. The member also reaches the default state with an administrative shutdown. For 6300 and 6400 series switches, the administrative state is enabled. Configurations, such as MTU and UDLD, are retained.

Command context

```
config-if
```

Parameters

<ID>

Specifies a LAG interface ID. Range: 1 to 256.

Authority

Administrators or local user group members with execution rights for this command.

Usage

- All members of the LAG must have the same speed. If a member comes up late with a different speed, it will not participate in the LAG/LACP. The hardware restriction is applied before adding an interface to LAG. The member belongs to the card type that has the same maximum speed as the reference port card type.
- To move an interface from LagA to LagB, first remove the interface from LagA and then add it to LagB. When a member is attached to a LAG, the nondefault configurations on the member are removed silently.
- After removing a physical interface from a LAG, the interface associated with the LAG becomes L3 ports with default L3 configurations and administrative down. For example, suppose interface 1/1/1 was part of LAG 3 and you had administratively enabled the interface. If you later remove interface 1/1/1 from LAG 3, the administrative status automatically changes to down. If you want to use the interface again, you must administratively enable it again.

Examples

Adding an interface to a Link Aggregation Group (LAG) represented by an ID of 100:

```
switch(config)# interface 1/1/1  
switch(config-if)# lag 100
```

Deleting an interface from a Link Aggregation Group (LAG) represented by an ID of 100:

```
switch(config)# interface 1/1/1  
switch(config-if)# no lag 100
```

show interface

Syntax

```
show interfaces <LAG-NAME> [vsx-peer]
```

Description

Displays information about a specific LAG.

Command context

Operator (>) or Manager (#)

Parameters

<LAG-NAME>

Specifies a LAG name.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying information about LAG 100:

```
switch# show interface lag100  
Aggregate lag100 is up  
Admin state is up  
Description :  
MAC Address      : 48:0f:cf:af:43:9c  
Aggregated-interfaces : 1/1/2  
Aggregation-key   : 100  
Aggregate mode    : active  
Speed             : 2000 Mb/s  
L3 Counters: Rx Disabled, Tx Disabled  
qos trust none  
VLAN Mode: access  
Access VLAN: 1  
Rx  
      20 input packets      2584 bytes  
      0 input error        0 dropped  
      0 CRC/FCS
```

Tx	45 output packets	5658 bytes
	0 input error	4 dropped
	0 collision	

show lacp aggregates

Syntax

```
show lacp aggregates [<LAG-NAME>] [vsx-peer]
```

Description

Displays all LACP aggregate information configured for all LAGs, or for a specific LAG.

Command context

Operator (>) or Manager (#)

Parameters

<LAG-NAME>

Optional: Specifies a lag name.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying LACP aggregate information configured for lag10:

```
switch# show lacp aggregates lag10
```

```
Aggregate-name       : lag10
Aggregated-interfaces : 1/1/1 1/1/2
Heartbeat rate       : slow
Hash                 : 13-src-dst
Aggregate mode        : active
```

Displaying LACP aggregates:

```
switch# show lacp aggregates
```

```
Aggregate-name       : lag1
Aggregated-interfaces : 1/1/27 1/1/28 1/1/29
Heartbeat rate       : slow
Hash                 : 13-src-dst
Aggregate mode        : active
```

```
Aggregate-name       : lag2
Aggregated-interfaces : 1/1/48
Heartbeat rate       : slow
```

```
Hash : l2-src-dst
Aggregate mode : passive
```

show lacp configuration

Syntax

```
show lacp configuration [vsx-peer]
```

Description

Displays global LACP configuration.

Command context

Operator (>) or Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying global LACP configuration (output is applicable for 8400 series switches):

```
switch# show lacp configuration
System-id      : 70:72:cf:ef:fc:d9
System-priority : 65534
Hash           : l3-src-dst
```

Displaying global LACP configuration (output is applicable for 8320, 6300, and 6400 series switches):

```
switch# show lacp configuration
System-id      : 98:f2:b3:68:40:a0
System-priority : 65534
```

show lacp interfaces

Syntax

```
show lacp interfaces [<IFNAME>] [vsx-peer]
```

Description

Displays an LACP configuration of the physical interfaces, including VSXs. If an interface name is passed as argument, it only displays an LACP configuration of a specified interface.

Command context

Operator (>) or Manager (#)

Parameters

<IFNAME>

Optional: Specifies an interface name.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

This example displays an LACP configuration of the physical interfaces. One of the interfaces has the `lacp-block` forwarding state. If a VSX switch has loop protect enabled on an interface and a loop occurs, VSX blocks the interface to stop the loop. The forwarding state of the blocked interface is set to `lacp-block`.

```
switch# show lacp interfaces
State abbreviations :
A - Active           P - Passive           F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync      O - OutofSync
C - Collecting       D - Distributing
X - State m/c expired E - Default neighbor state
```

Actor details of all interfaces:

Intf	Aggr name	Port id	Port Pri	State	System-id	System Pri	Aggr Key	Forwarding State
1/1/1	lag10	17	1	ALFOE	70:72:cf:37:a3:5c	20	10	lacp-block
1/1/2	lag128	69	1	ALFNCD	70:72:cf:37:a3:5c	20	128	up
1/1/3	lag128	14	1	ALFNCD	70:72:cf:37:a3:5c	20	128	up
1/1/4	lag128							down
1/1/5	lag20							up

Partner details of all interfaces:

Intf	Aggr name	Partner Port-id	Port Pri	State	System-id	System Priority	Aggr Key
1/1/1	lag10	0	65534	PLFOEX	00:00:00:00:00:00	65534	0
1/1/2	lag128	69	1	PLFNCD	70:72:cf:8c:60:a7	65534	128
1/1/3	lag128	14	1	PLFNCD	70:72:cf:8c:60:a7	65534	128
1/1/4	lag128						
1/1/5	lag20						

Displaying static LAG:

```
switch# show lacp interfaces
State abbreviations :
A - Active           P - Passive           F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync      O - OutofSync
C - Collecting       D - Distributing
X - State m/c expired E - Default neighbor state
```

Actor details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-id	System Pri	Aggr Key	Forwarding State
------	-----------	---------	----------	-------	-----------	------------	----------	------------------

```

-----
1/1/1 lag10 up
1/1/2 lag10 up

```

Partner details of all interfaces:

```

-----
Intf    Aggr    Port    Port    State    System-id    System    Aggr
      Name    Id      Pri
-----
1/1/1 lag10
1/1/2 lag10

```

Displaying an LACP configuration of the 1/1/1 interface:

```
switch# show lacp interfaces 1/1/1
```

```

State abbreviations :
A - Active           P - Passive           F - Aggregable I - Individual
S - Short-timeout    L - Long-timeout    N - InSync      O - OutofSync
C - Collecting       D - Distributing
X - State m/c expired           E - Default neighbor state

```

Aggregate-name : lag1

```

-----
Actor                                Partner
-----
Port-id          | 28          | 31
Port-priority    | 1           | 1
Key              | 1           | 1
State            | ALFNCD      | ALFNCD
System-id        | 98:f2:b3:68:40:a0 | 98:f2:b3:68:60:a6
System-priority  | 65534       | 65534

```

Displaying an LACP configuration after loop-protect is enabled on the primary VSX switch:

```
switch# show lacp interfaces
```

```

State abbreviations :
A - Active           P - Passive           F - Aggregable I - Individual
S - Short-timeout    L - Long-timeout    N - InSync      O - OutofSync
C - Collecting       D - Distributing
X - State m/c expired           E - Default neighbor state

```

Actor details of all interfaces:

```

-----
Intf    Aggr    Port    Port    State    System-ID    System    Aggr    Forwarding
      Name    Id      Pri
-----
1/4/14 lag1(mc) 206    1      ALFNCD  f8:60:f0:06:49:00 65534  1      up
1/5/15 lag2(mc)                down

```

Partner details of all interfaces:

```

-----
Intf    Aggr    Port    Port    State    System-ID    System    Aggr
      Name    Id      Pri
-----
1/4/14 lag1(mc) 130    1      ALFNCD  f8:60:f0:06:87:00 65534  1
1/5/15 lag2(mc)

```

Displaying an LACP configuration after loop-protect is enabled on the secondary VSX switch:

```
switch# show lacp interfaces
```

State abbreviations :

A - Active P - Passive F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync O - OutofSync
C - Collecting D - Distributing
X - State m/c expired E - Default neighbor state

Actor details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-ID	System Pri	Aggr Key	Forwarding State
1/3/2	lag1 (mc)	1130	1	ALFNCD	f8:60:f0:06:49:00	65534	1	up
1/9/3	lag2 (mc)							down

Partner details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-ID	System Pri	Aggr Key
1/3/2	lag1 (mc)	131	1	ALFNCD	f8:60:f0:06:87:00	65534	1
1/9/3	lag2 (mc)						

shutdown

Syntax

```
shutdown
```

```
no shutdown
```

Description

Sets every interface in the LAG operationally down.

The `no` form of this command sets every interface operationally up.

Command context

```
config-lag-if
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting every interface in the LAG to shutdown:

```
switch(config)# interface lag 1  
switch(config-lag-if)# shutdown
```

Resetting every interface in the LAG to the default (up):

```
switch(config)# interface lag 1  
switch(config-lag-if)# no shutdown
```


vlan trunk native

Syntax

```
vlan trunk native <VLAN-ID>
```

```
no vlan trunk native [<VLAN-ID>]
```

Description

Assigns a native VLAN ID to a LAG interface.

The `no` form of this command removes a native VLAN from a LAG interface and assigns VLAN ID 1 as its native VLAN.

Command context

```
config-lag-if
```

Command context

```
config-if
```

Parameters

<VLAN-ID>

Specifies the number of the VLAN ID to assign. The VLAN ID must exist.

Maximum number of VLANs supported: 512 (6100); 2048 (6200); 4096 (6300, 6400, 8320, 8325, 8360, 8400).

VLAN ID range: 2 to 4094.

Authority

Administrators or local user group members with execution rights for this command.

Usage

By default, VLAN ID 1 is assigned as the LAG VLAN ID for all LAG interfaces. VLANs can only be assigned to a nonrouted (layer 2) interface or LAG interface.

Only one VLAN ID can be assigned as the native VLAN. For the interface to forward the native VLAN traffic, the interface has to be allowed explicitly by entering `vlan trunk allowed <ID>` where the ID is the native VLAN ID. This setting is also applicable to the physical interface.

Examples

Configuring a Layer 2 dynamic aggregation group with native VLAN ID 1 assigned to LAG 1:

For 6100 and 6200 switch series:

```
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# vlan trunk native 1
switch(config-lag-if)# vlan trunk allowed 1
```

For 6300, 6400, 8320, 8325, 8360, and 8400 switch series:

```
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
switch(config-lag-if)# no routing
switch(config-lag-if)# lacp mode active
```

```
switch(config-lag-if) # vlan trunk native 1  
switch(config-lag-if) # vlan trunk allowed 1
```

Configuring a Layer 2 dynamic aggregation group with native VLAN ID 20 assigned to LAG 1:

For 6100 and 6200 switch series:

```
switch(config) # interface lag 1  
switch(config-lag-if) # no shutdown  
switch(config-lag-if) # lacp mode active  
switch(config-lag-if) # vlan trunk native 20  
switch(config-lag-if) # vlan trunk allowed 20
```

For 6300, 6400, 8320, 8325, 8360, and 8400 switch series:

```
switch(config) # interface lag 1  
switch(config-lag-if) # no shutdown  
switch(config-lag-if) # no routing  
switch(config-lag-if) # lacp mode active  
switch(config-lag-if) # vlan trunk native 20  
switch(config-lag-if) # vlan trunk allowed 20
```

Removing a native VLAN from LAG 1:

```
switch(config) # interface lag 1  
switch(config-lag-if) # no vlan trunk native
```

Accessing Aruba Support

Aruba Support Services	https://www.arubanetworks.com/support-services/
Aruba Support Portal	https://asp.arubanetworks.com/
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
Software licensing	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/
Aruba software and documentation	https://asp.arubanetworks.com/downloads

Accessing updates

To download product updates:

Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>



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Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

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